

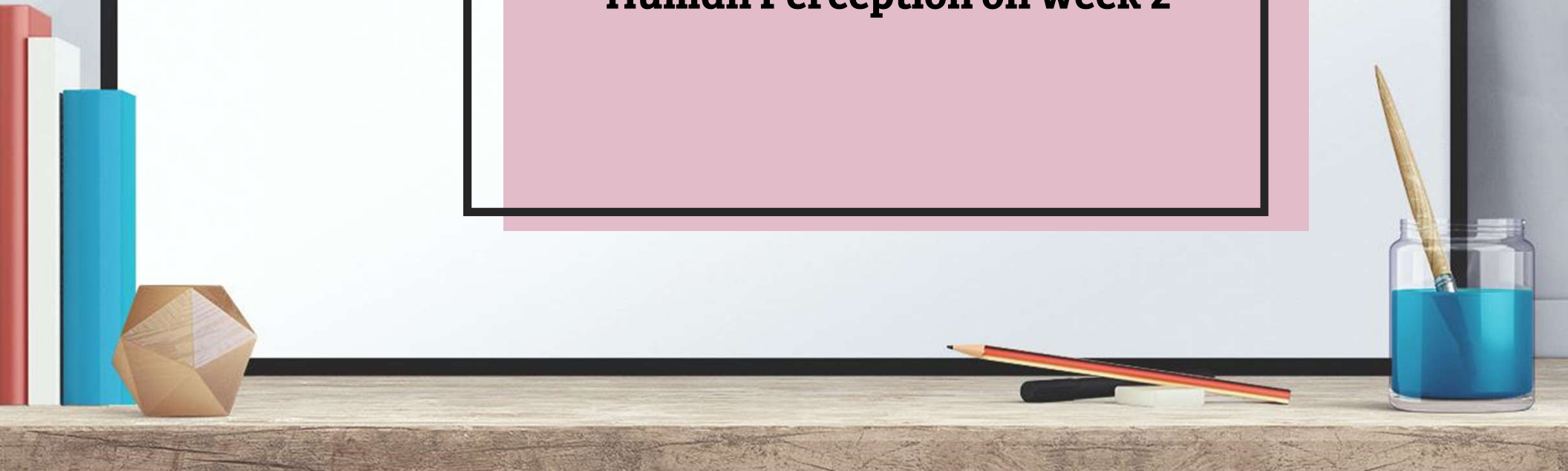
HCI Week2: Waiting for your friends coming into this class

<https://www.youtube.com/watch?v=HvuoCNv9ufk>



Human & Computer Interaction

Human Perception on week 2



Week 2 - Human Perception Agenda

- What is Human Perception?
- Visual Perception
- Hearing Perception
- Touch Perception
- Taste Perception
- Olfactory Perception

WHY DO YOU NEED TO STUDY HUMAN PERCEPTION?





HUMAN AND COMPUTER INTERACTION

Cognitive process

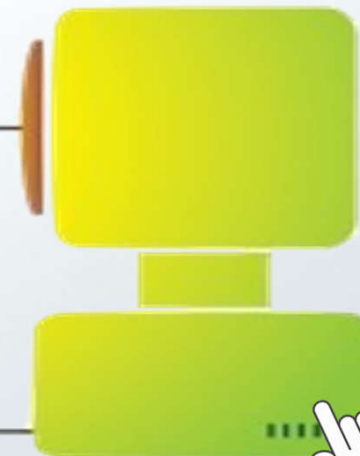
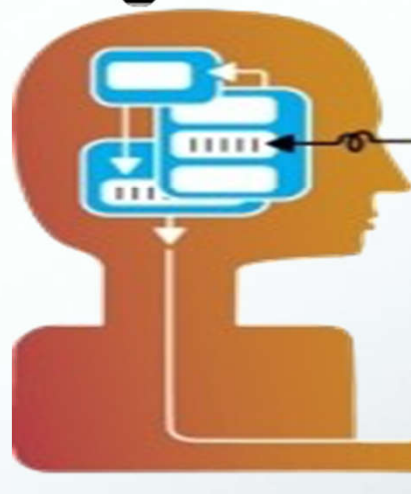
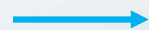


1. ช่องทางการรับเข้าออกข้อมูล
2. ความจำของมนุษย์
3. การประมวลผลของมนุษย์ เช่น การใช้เหตุผล การตัดสินใจ

Input

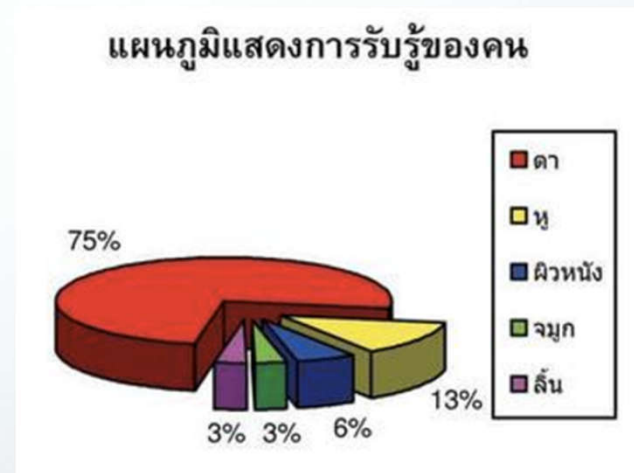
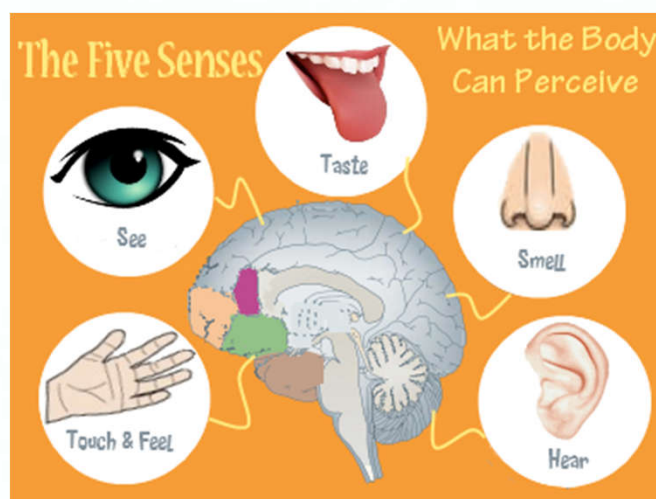


Output





HUMAN PERCEPTION

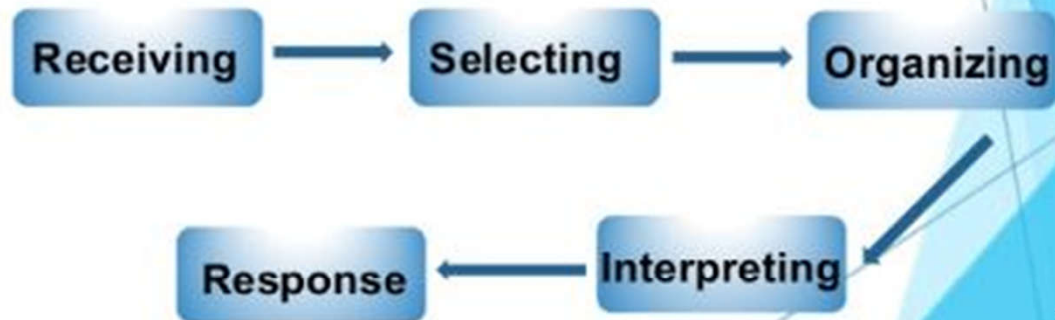


Eye
Ear
Skin
Nose
Tongue



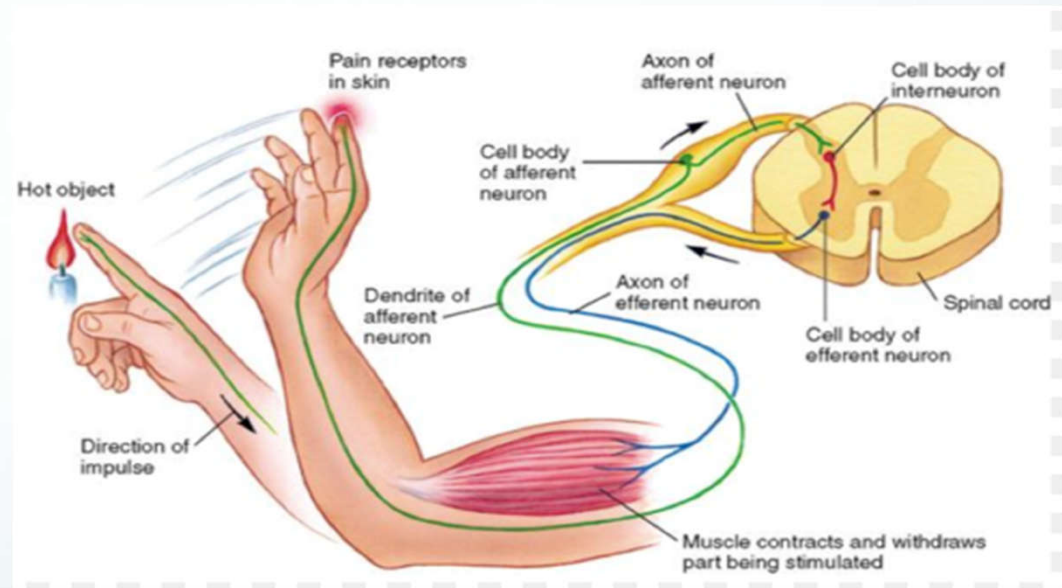
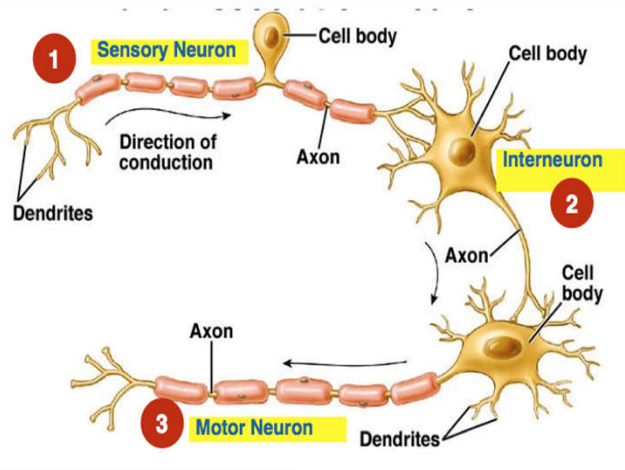
PERCEPTUAL PROCESS

- ▶ A process by which individuals organize and interpret their sensory impressions in order to give meaning to their environment.



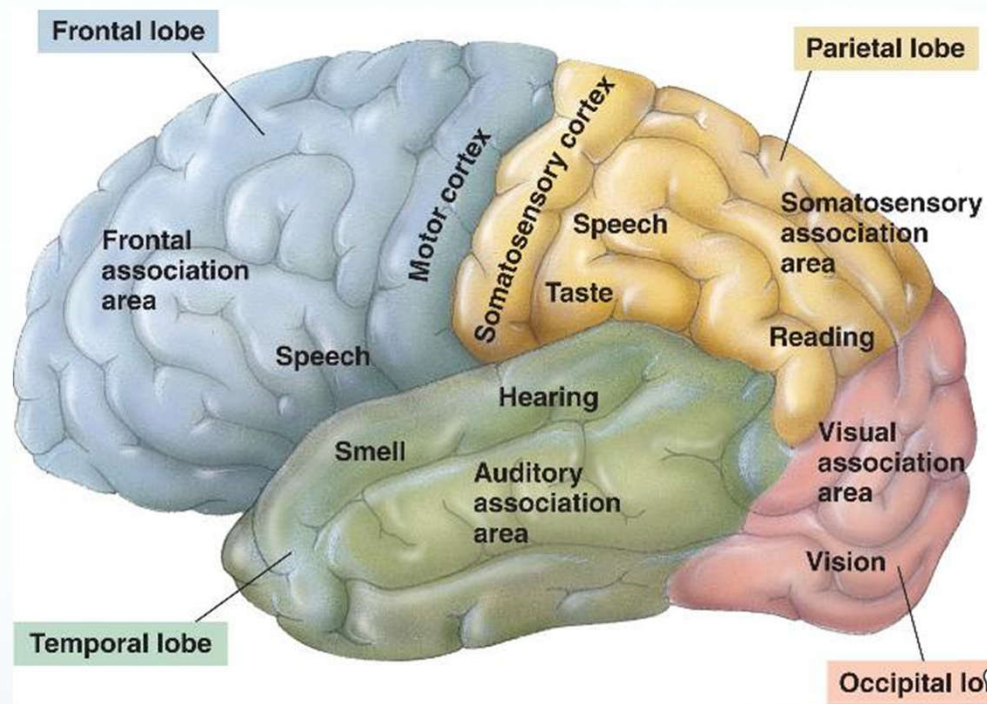


THE MECHANISM OF HUMAN PERCEPTION





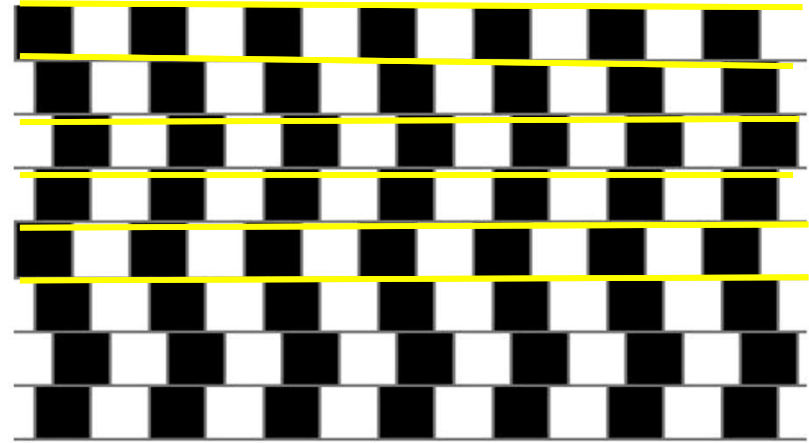
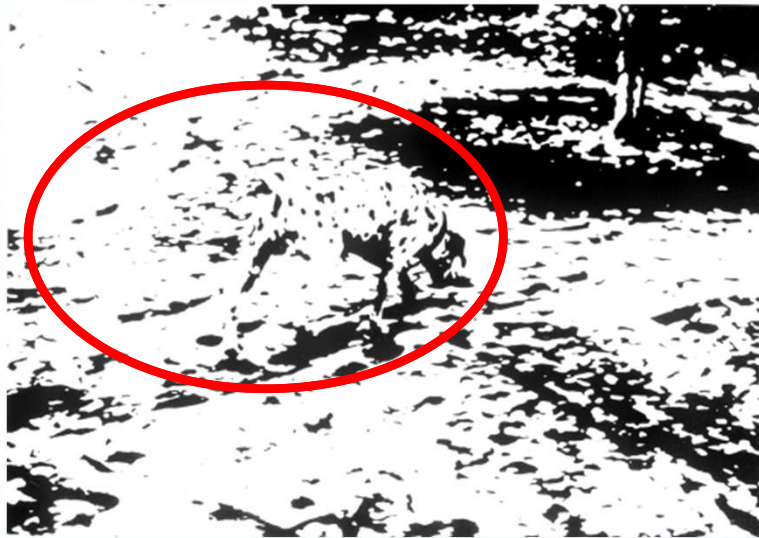
HUMAN BRAIN



<http://bio1152.nicerweb.com/Locked/media/ch49/brain-lobes.html>

What do you see here?

WHAT DO YOU SEE HERE?



Are the horizontal lines parallel or do they slope?

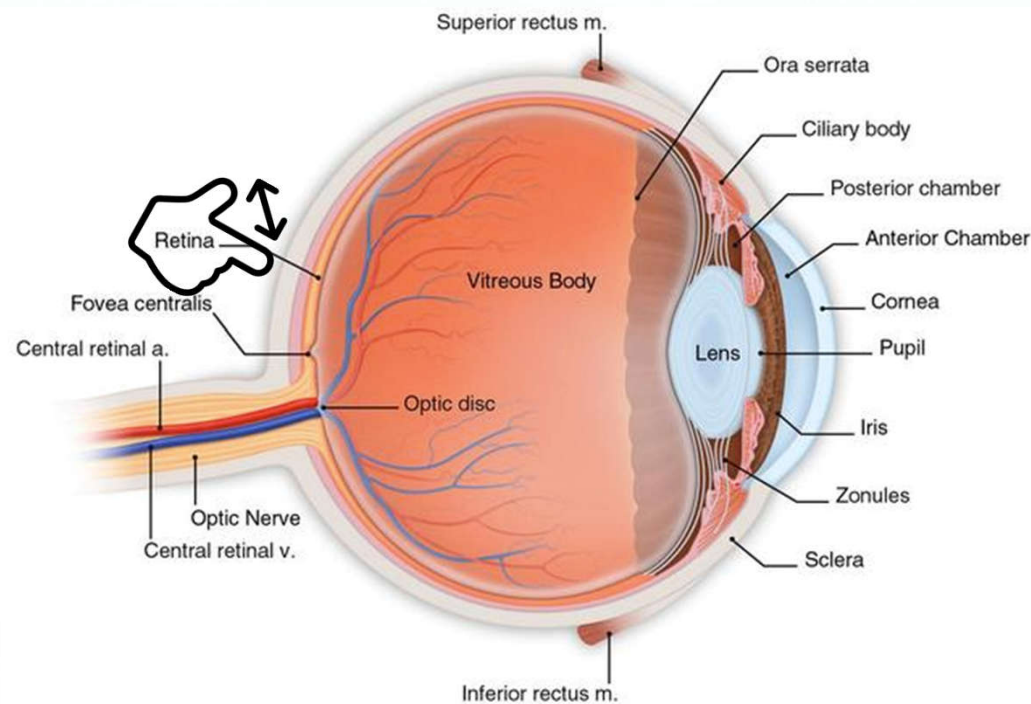
Students, write your response!

<http://www.michaelbach.de/ot/index.html>

Pear Deck Interactive Slide
Do not remove this bar

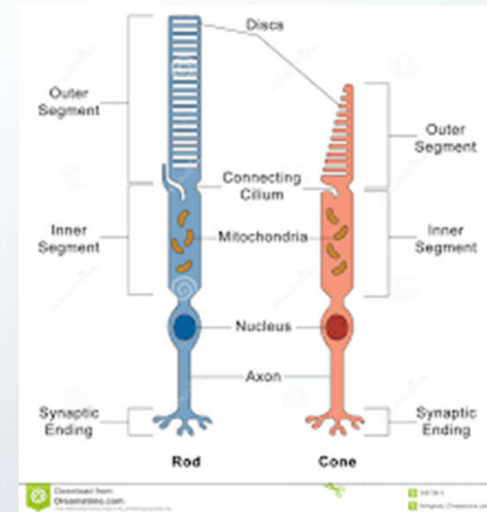
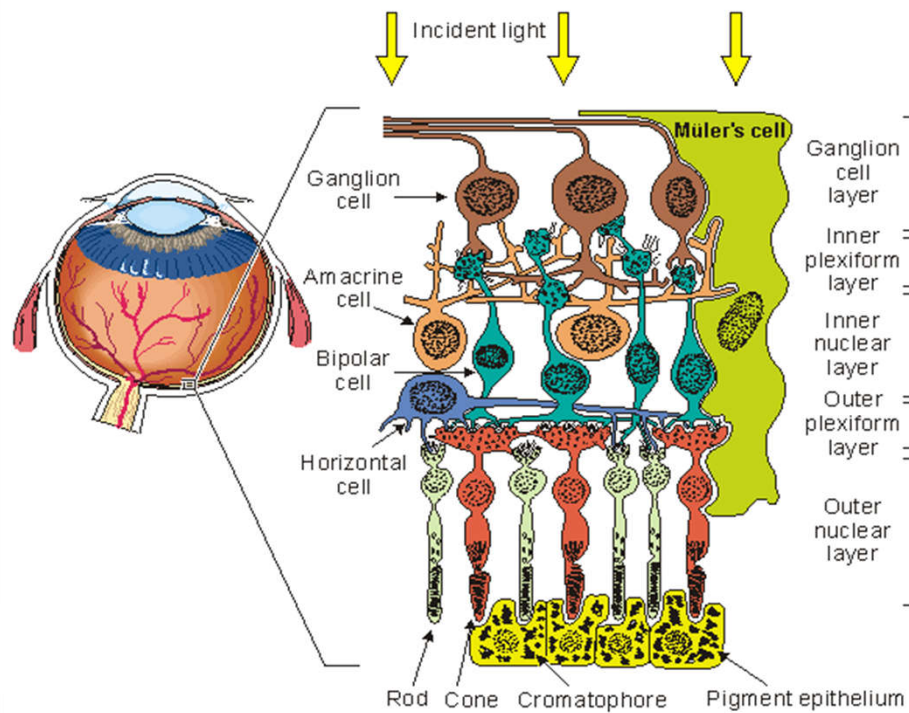


HUMAN EYE ANATOMY



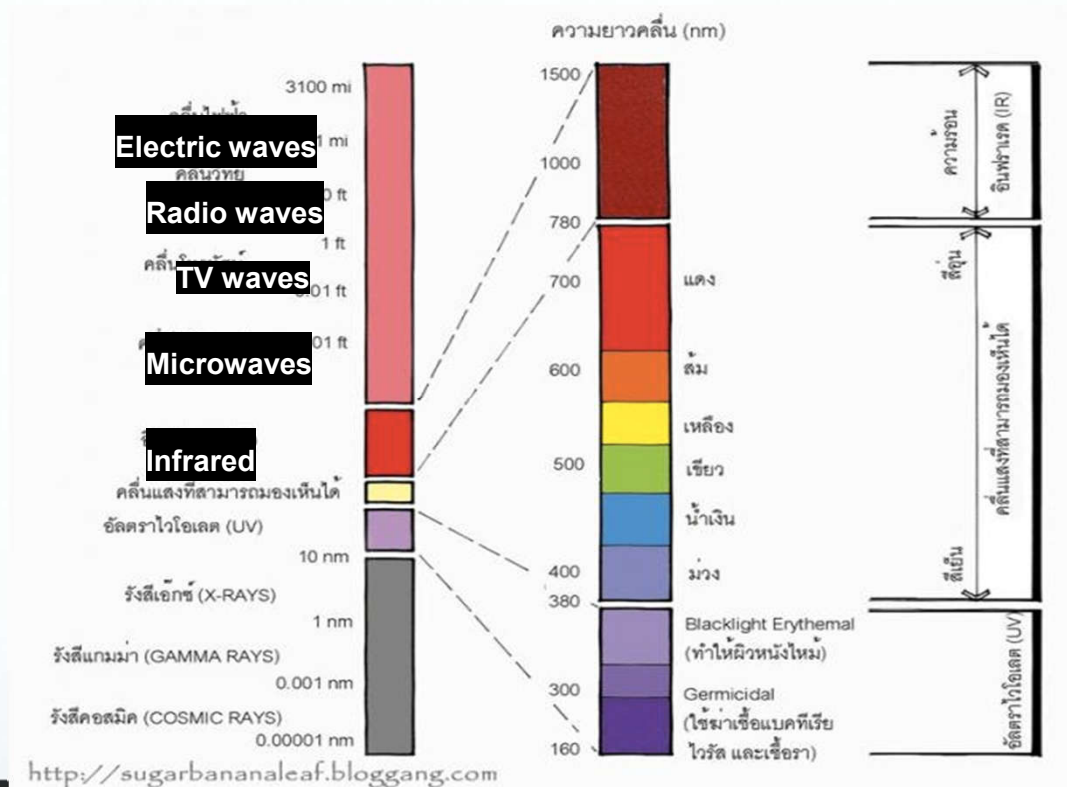
http://www.midwestglaucoma.com/eyes_structure.html

CELLS IN RETINA





VISIBLE SPECTRUM





HUMAN VS ANIMAL EYES TEST

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=ISY_XSTCCKW](https://www.youtube.com/watch?v=ISY_XSTCCKW)



<https://knowledge.jarm.com/view/67117>



WHAT IS THE ANIMAL VISION?



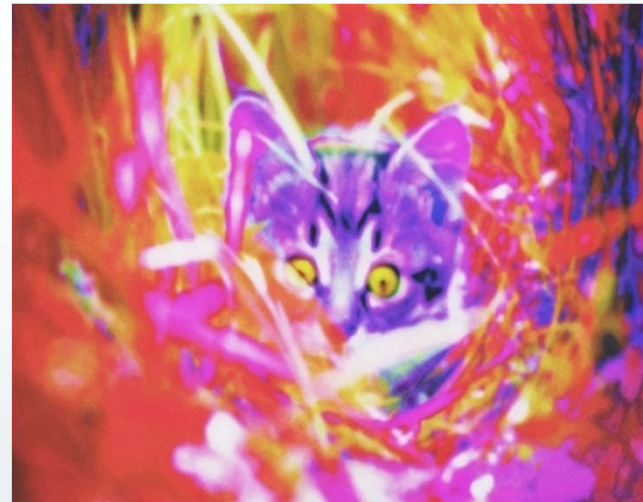
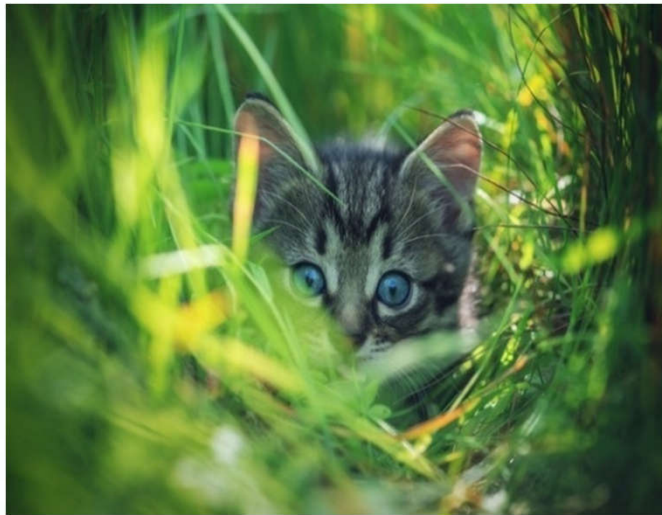
Students, write your response!

<https://knowledge.jarm.com/view/67117>

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WHAT IS THE ANIMAL VISION?



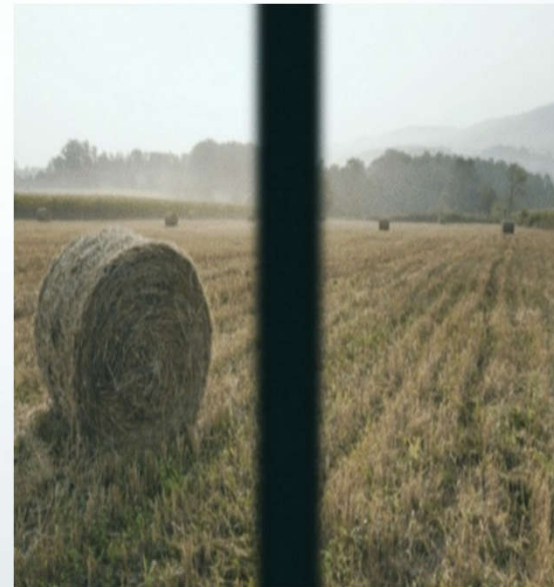
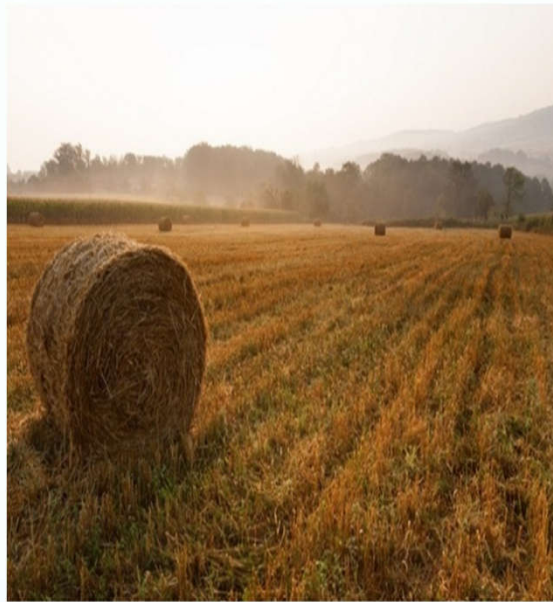
Students, write your response!

<https://knowledge.jarm.com/view/67117>

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Students, write your response!

<https://knowledge.jarm.com/view/67117>

Pear Deck Interactive Slide
Do not remove this bar



WHAT IS THE ANIMAL VISION?



Students, write your response!

<https://knowledge.jarm.com/view/67117>

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THE VISUAL CLIFF STUDY

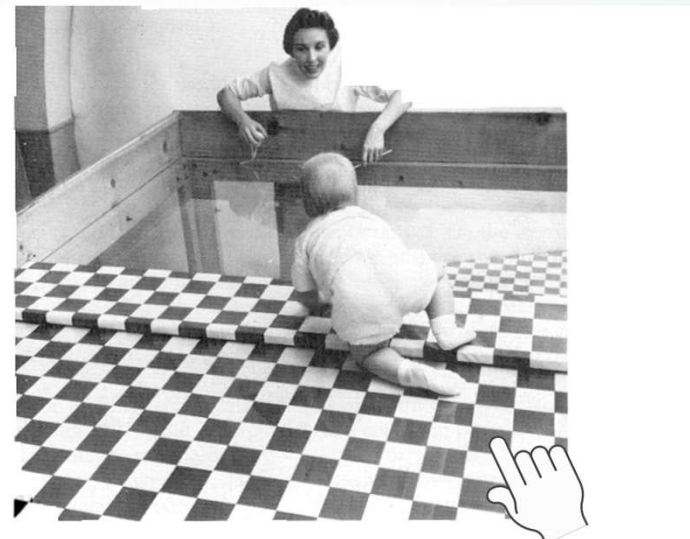
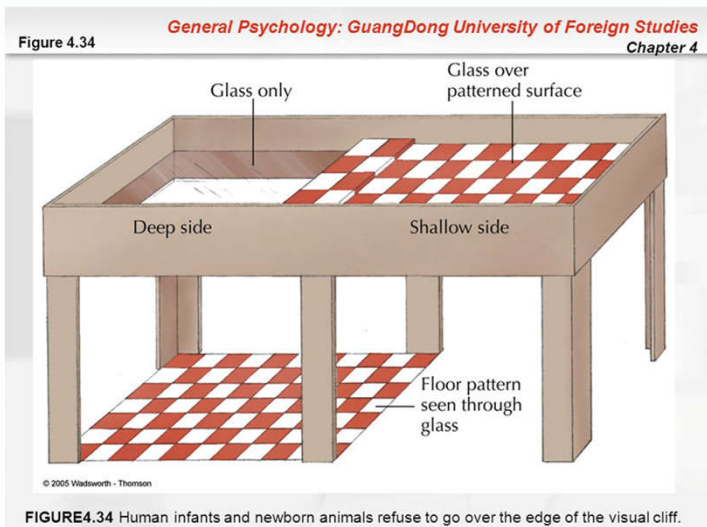


Figure 12.12 A mother calling to her child from across the deep side of the visual cliff. Despite the presence of the glass surface covering the cliff, the child refuses to cross over to the mother. (From Gibson & Walk, 1960. Photograph courtesy of William Vandivert.)

THESE FACTORS HELP HUMAN PERCEIVE DEPTH

Size



Interposition



Texture



Perceptive

DEPTH IN INTERFACE DESIGN

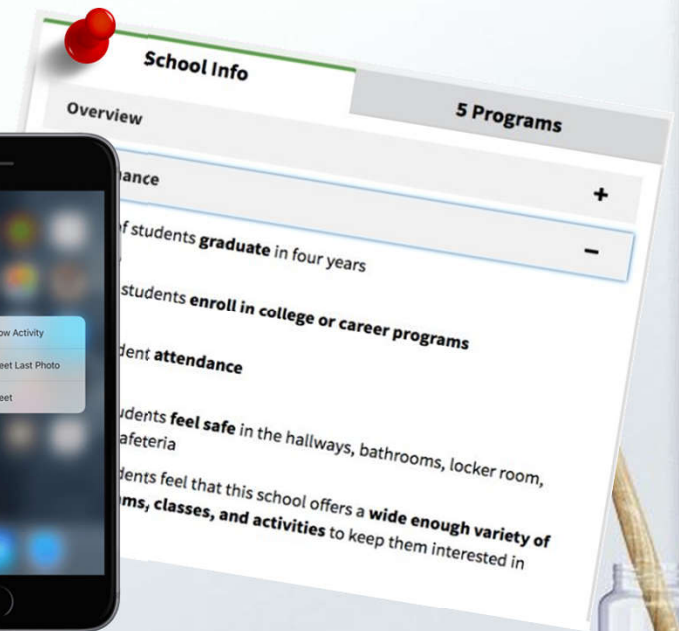
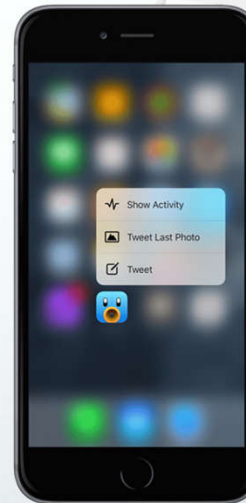
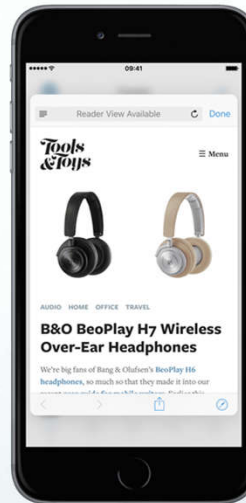
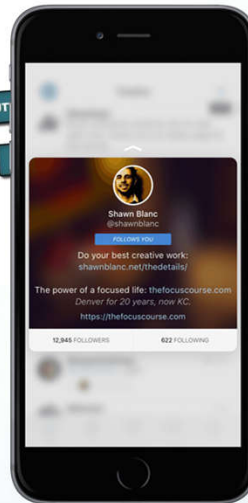
FEW AFFORDANCES

I'M A BUTTON

I'M A BUTTON

I'M A BUTTON

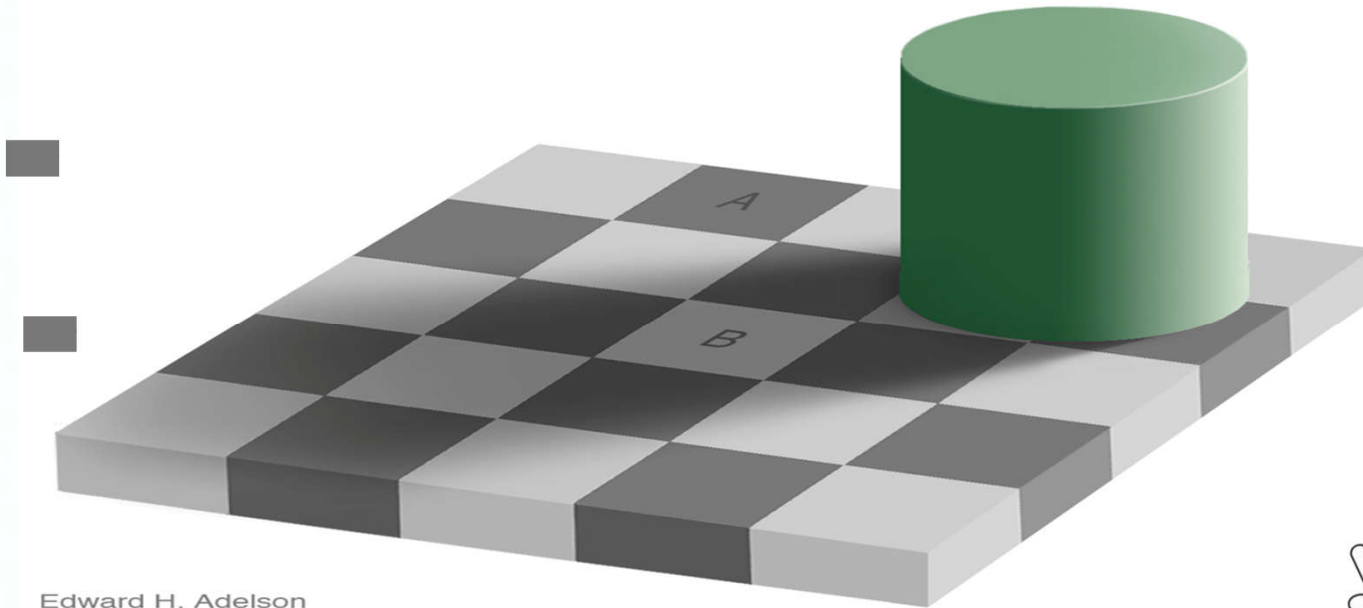
I'M A BUT



<https://justuxdesign.com/blog/what-are-affordances-in-ui-design-and-why-do-they-matter>

<https://www.smashingmagazine.com/2017/06/designing-perfect-accessibility-checklist/>

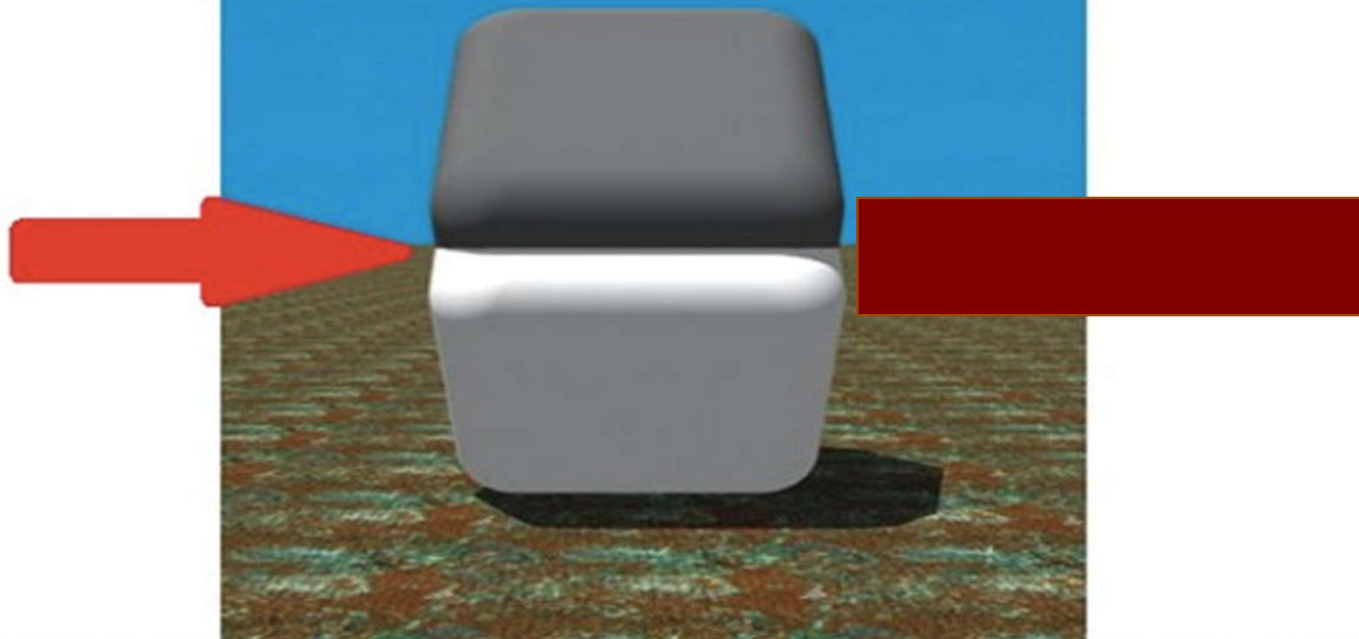
THE CORNSWEET ILLUSION



Edward H. Adelson

Due to the arrangement of images, effect of colors, impact of light source or other variable, a wide range of misleading visual effects can be seen.

VISUAL ILLUSION



<http://www.michaelbach.de/ot/index.html>

VISUAL PERCEPTUAL

คำขวัญ
โรงเรียน
วัดหนองไผ่

ถ้าอ่านไม่ออก ลองทอยห่างออกไปเรื่อย ๆ



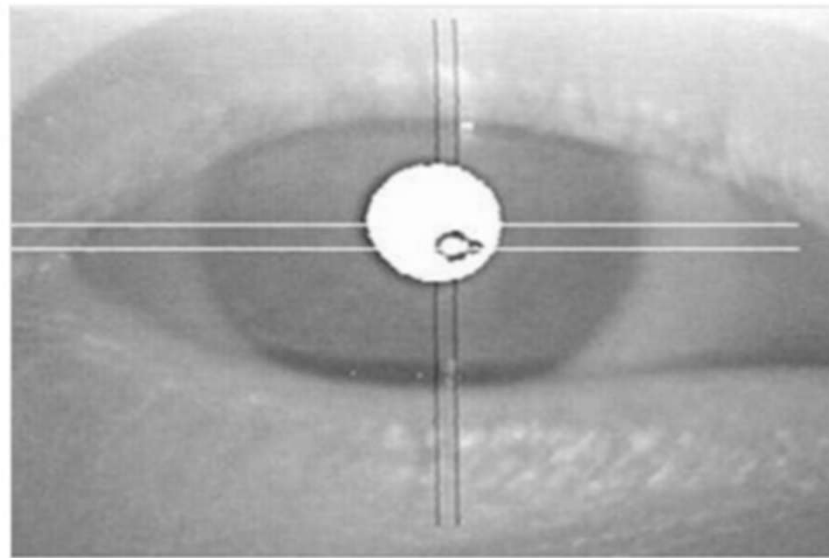
HOW DO WE SEE THE WORLD?



EYE TRACKING

The eye tracking monitor uses infrared light to **make the pupils of the person sitting in front of it light up** and so become detectable.

This is what it looks like for the monitor.



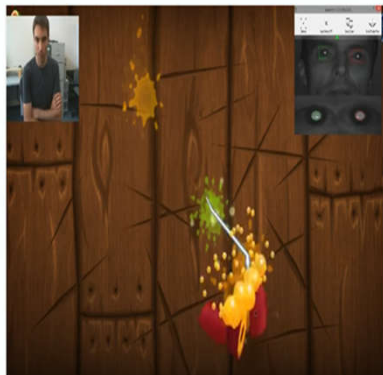
tobii

OptimalUsability

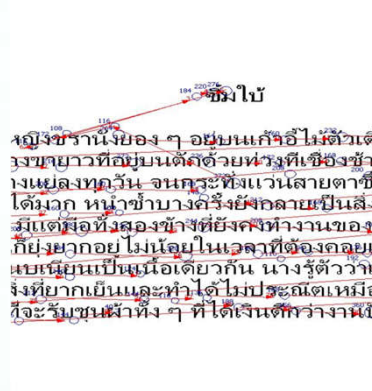
EYE TRACKERS



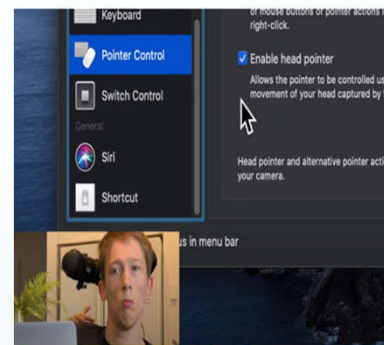
EYE TRACKING TECHNOLOGY



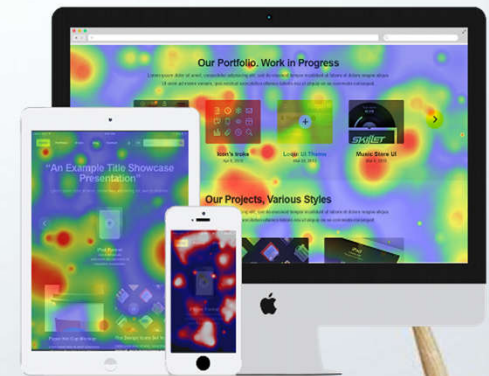
Game



Reading



Helping disabled people



UX research

SOUND

How do we hear and speak?



HEARING PROCESS

How We Hear

1.

Sound waves enter your outer ear and travel through the ear canal to your eardrum.

2.

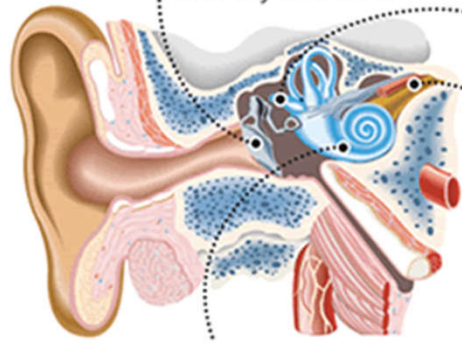
Your eardrum vibrates with the incoming sound and sends the vibrations to three tiny bones in your middle ear.

3.

The bones in your middle ear amplify the sound vibrations and send them to your inner ear, or cochlea. The sound vibrations activate tiny hair cells in the inner ear, which in turn release neurochemical messengers.

4.

Your auditory nerve carries this electrical signal to the brain, which translates it into a sound you can understand.

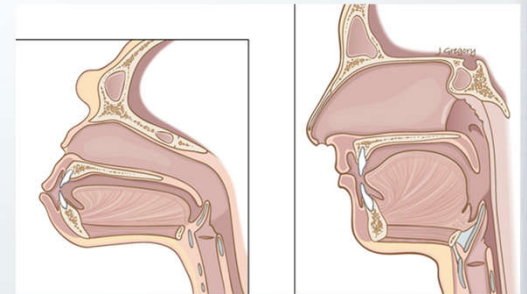


CARD00377-00-EE-ST-SL 10/09

SPEECH ORGAN

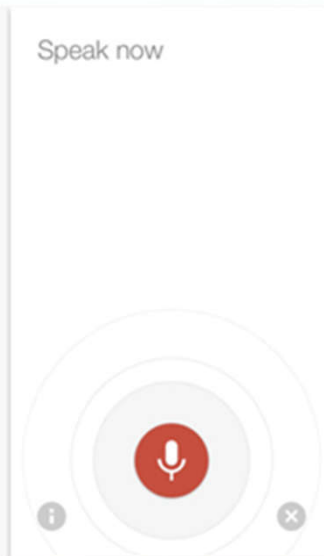
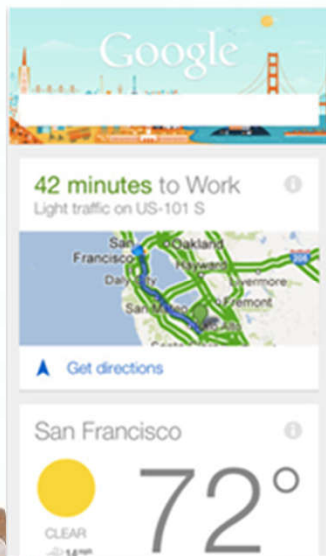
Richard Leakey explains what it is about the human body that enables speech. (*The Origin of Humankind*, 1994)

- Upright teeth in humans (fricative sounds)
- Intricate lip muscles on humans, flexible tongue, and small mouth
- Humans can close off the nose to create more air pressure for sound in the oral cavity
- The human larynx, or “voice box,” which contains the vocal cords, is in a lower position in humans than in other primates, creating a longer cavity called the pharynx



A VOICE USER INTERFACE

Can you tell me advantages and disadvantages of using voice in UI design?



MCGURK EFFECT

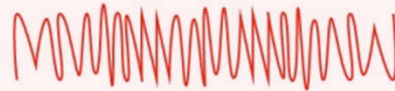
<https://www.youtube.com/watch?v=2k8fHR9jKVM>



BRAIN WAVE

BETA 14- 30 Hertz

(กำลังรวมการรับรู้ หลางใจ ไร้สติ)



โหมดชีวิตประจำวัน

ตื่นตัว กลัว ประใจ ตกใจ
จิตตก สับสน วุ่นวาย โกรธ

Baan Nada

www.baannada.com

www.facebook.com/baannada

ALPHA 9-13 Hertz

(ช่วงของการเรียนรู้ ปิงปอง ครีเอต)



นึกฝัน เริ่มมีสมาธิ สบายๆ

จิตใจ
สงบ

THETA 4-8 Hertz

(ช่วงที่เข้าสู่ภวังค์ มีความเชื่อใจกับการเห็นภาพต่างๆ)



หลับฝัน

สมาธิแบบลึก

DELTA 1-3 Hertz

(ช่วงที่เข้าสู่ฝันได้ลึกๆ เข้าถึงปัญญาระดับพิเศษได้)

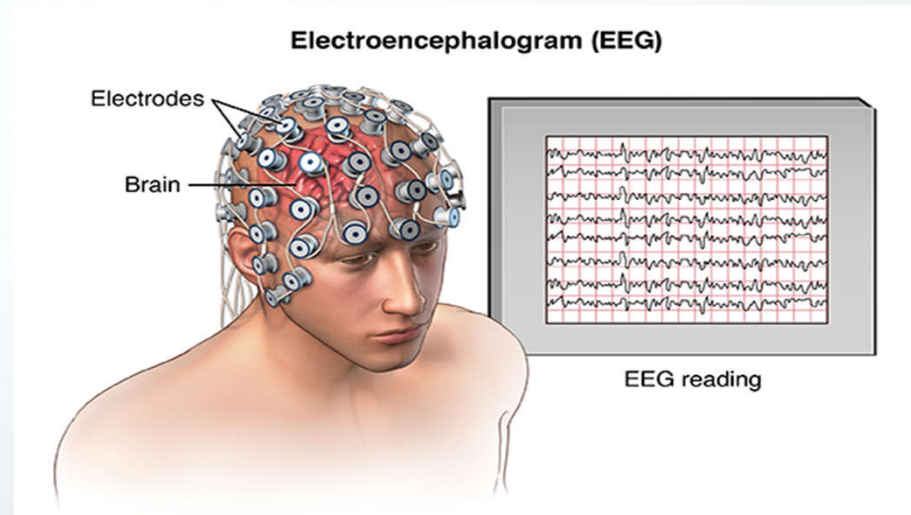
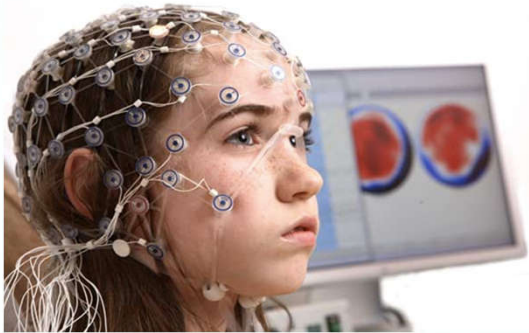


ไร้ตัวตน

เจ้านาสมาธิ
หลับลึกมากๆ



MIND CONTROL: EEG (ELECTROENCEPHALOGRAPH)



MIND CONTROL: EEG (ELECTROENCEPHALOGRAM)

<https://www.youtube.com/watch?v=DBYY3D1gkQ0>

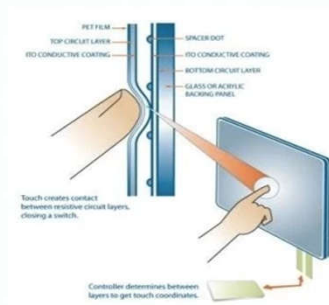
<http://www.olavkrigolson.com/that-neuroscience-guy/archives/04-2016>

TOUCH

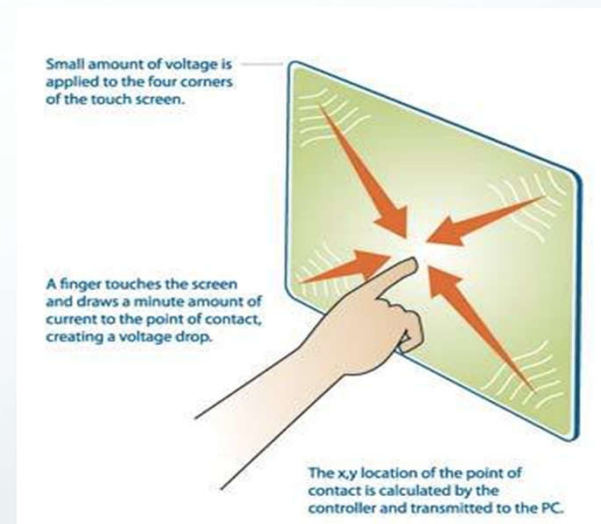
- A touch screen is a computer display screen that is sensitive to human touch, allowing a user to interact with the computer by touching pictures or words on the screen.
- At present there are many digital devices use sense of touch to control the interface.



TYPES OF TOUCH SCREEN



Resistive



Capacitive

HOW WE HOLD OUR SMARTPHONE



49%

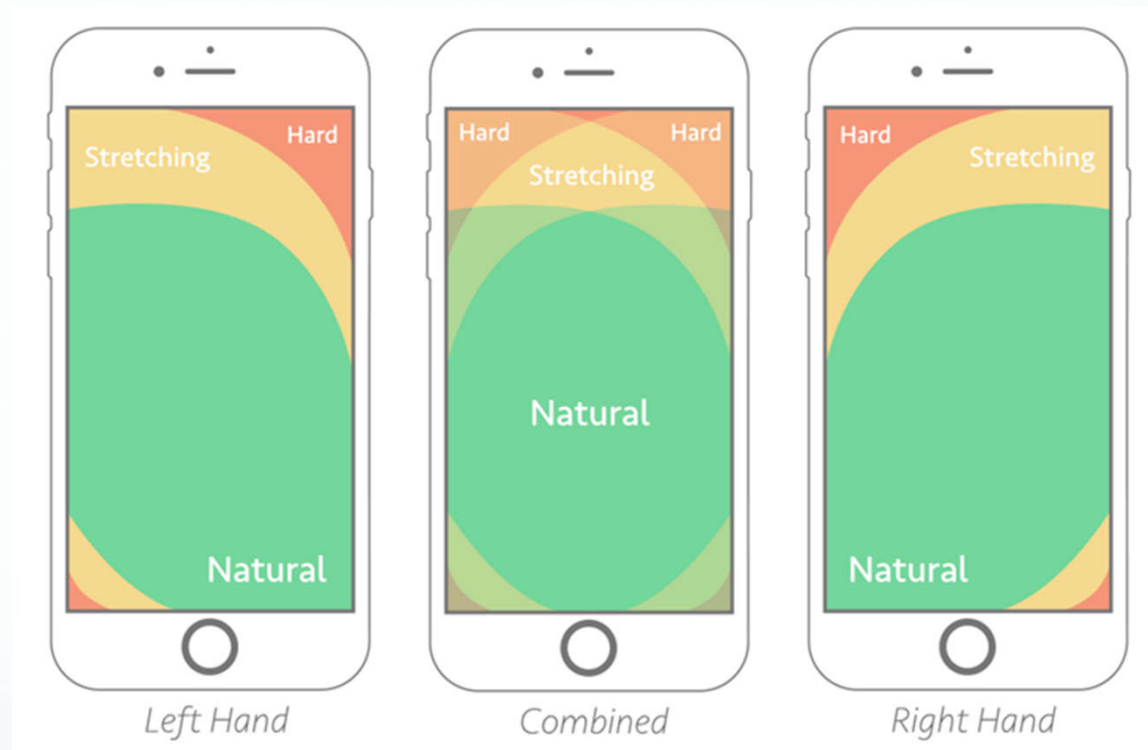


36%



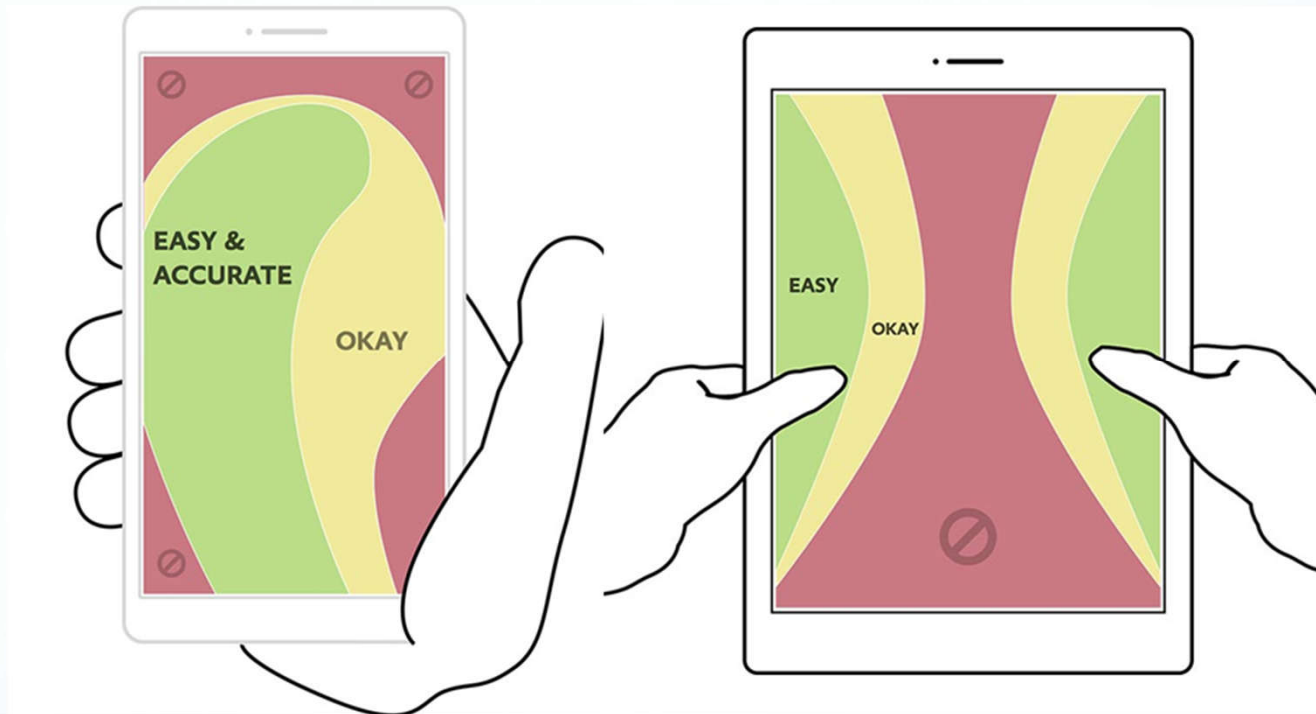
15%

THE THUMB ZONE: DESIGNING FOR MOBILE USERS



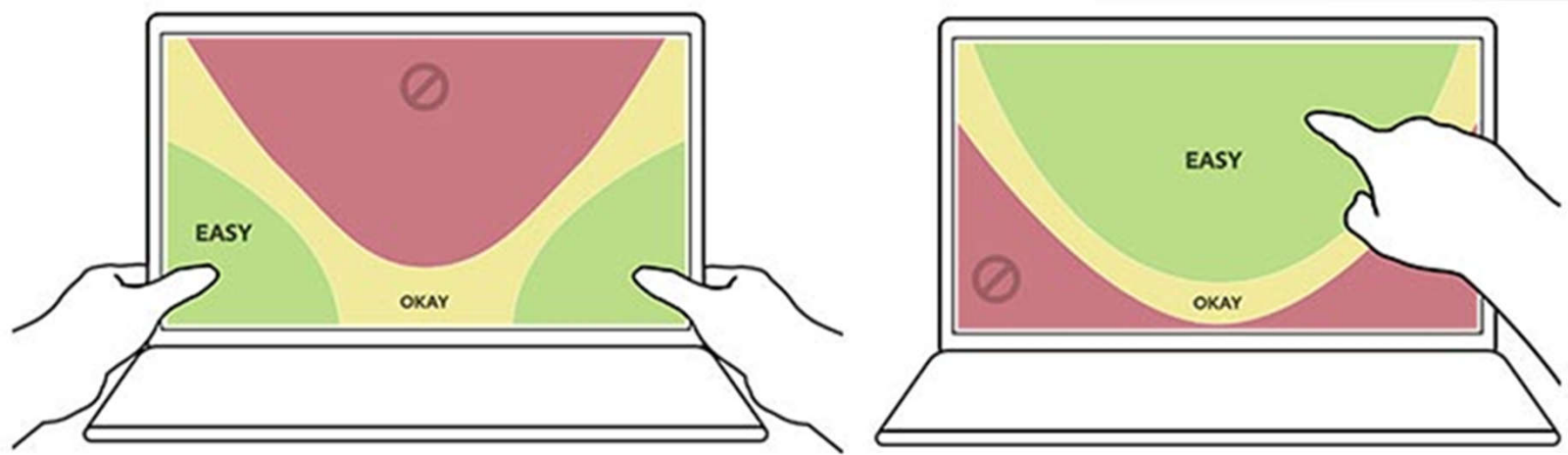
<https://www.smashingmagazine.com/2016/09/the-thumb-zone-designing-for-mobile-users/>

THE THUMB ZONE: DESIGNING FOR MOBILE USERS



<https://www.smashingmagazine.com/2016/09/the-thumb-zone-designing-for-mobile-users/>

THE THUMB ZONE: DESIGNING FOR MOBILE USERS

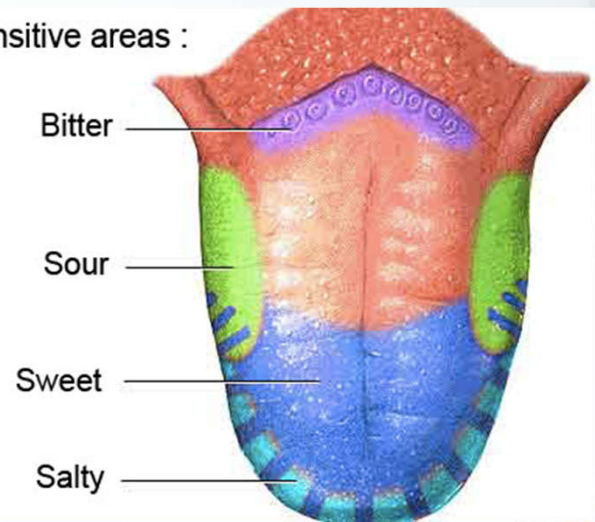


<https://www.smashingmagazine.com/2016/09/the-thumb-zone-designing-for-mobile-users/>

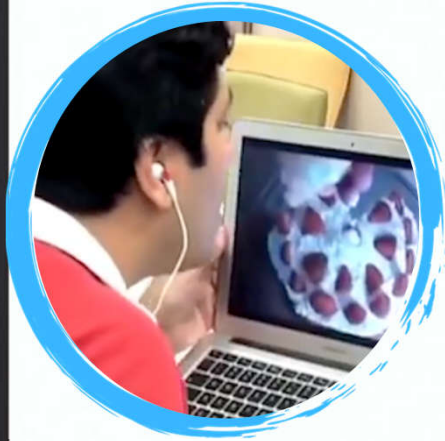
TASTE



Taste sensitive areas :



TASTE

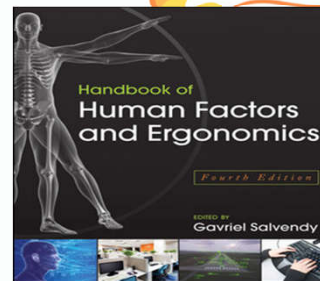


SMELL

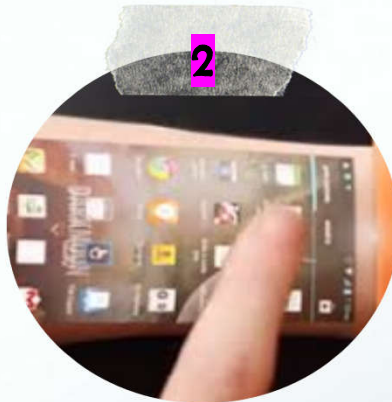
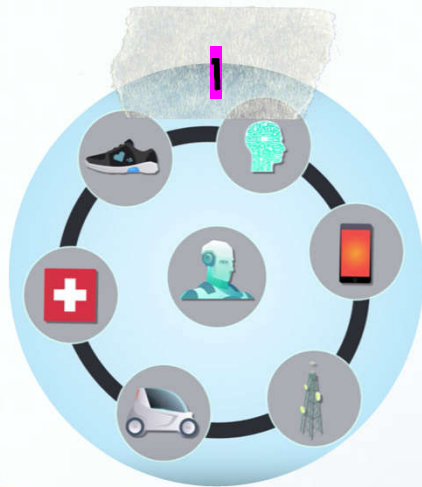


Week 3

Human factors & Ergonomics



FUTURE TECHNOLOGY (IDEA)



Students choose an option